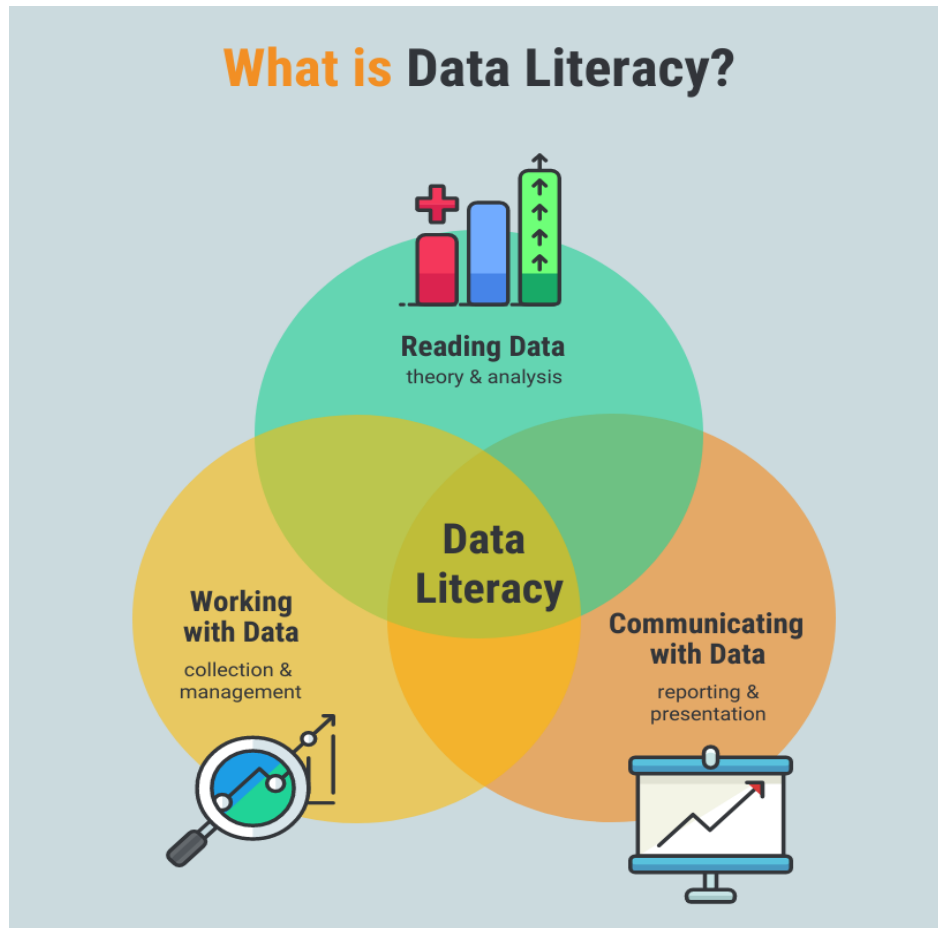


Short Answered Questions:

1Q. What is data literacy?

Data literacy is the ability to read, understand, analyze, work with, and communicate data effectively.



2Q. Explain why data literacy is becoming increasingly important in digital age.
(or)

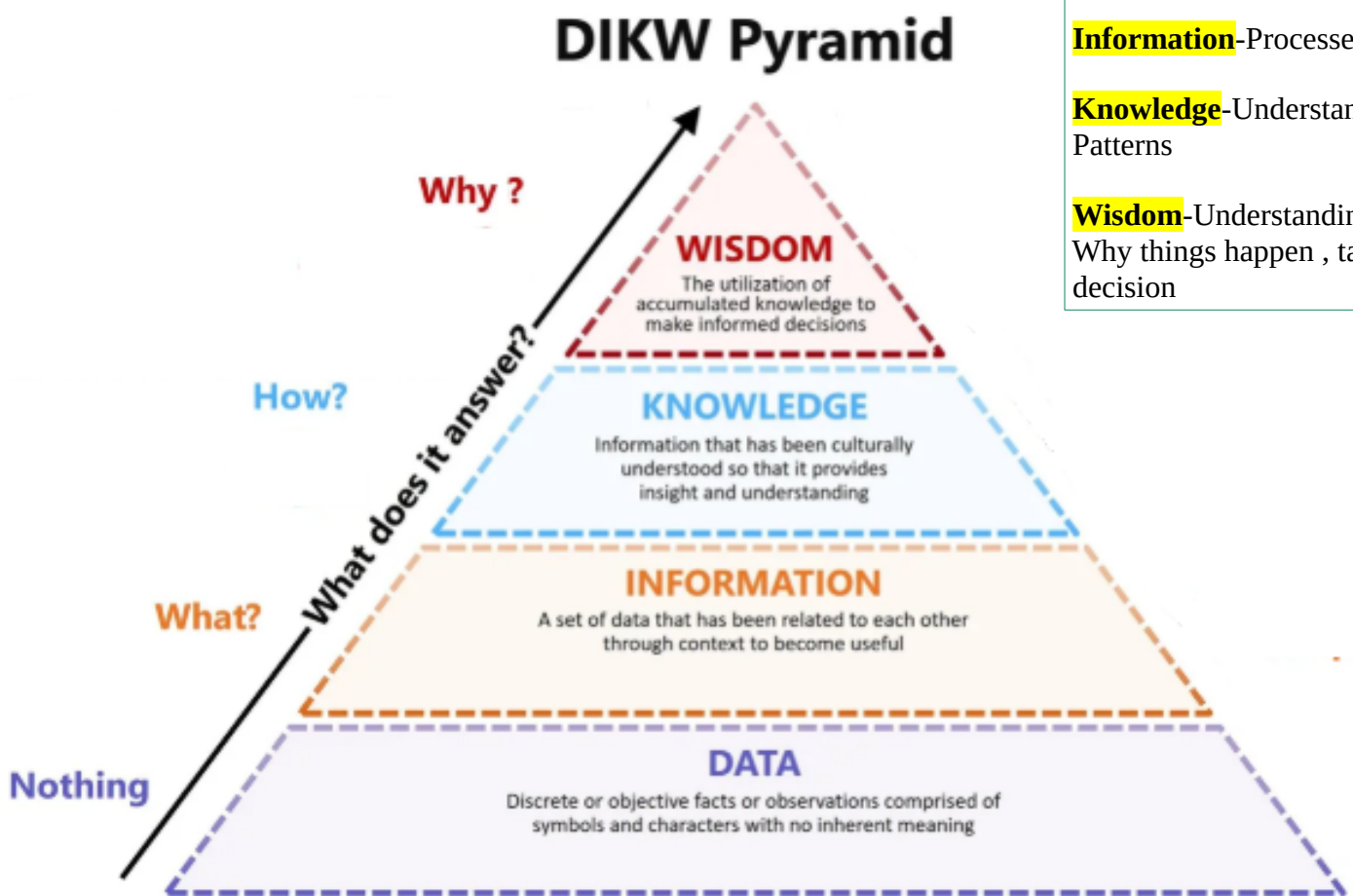
Q. List the benefits of being data literate.

Here are the main benefits of data literacy explained below:

1. **Better Decision Making** – Helps people understand facts and figures before taking decisions.
2. **Problem Solving** – Makes it easier to analyze problems and find solutions using data.
3. **Improved Communication** – People can present data clearly through charts, graphs, and reports.
4. **Higher Productivity** – Saves time because decisions are based on evidence, not guesswork.

3Q. What is the Data Pyramid?

- A data pyramid is also known as the **DIKW pyramid**.
- It represents the hierarchical relationship between **data**, **information**, **knowledge**, and **wisdom**.
- The pyramid gets narrower as we go up because each layer is rarer than the one below it: lots of raw data exists, but only some of it becomes useful information, even less becomes real knowledge, and true wisdom---knowing the right thing to do---is the rarest of all.



Data- Raw form

Information-Processed data

Knowledge-Understanding Patterns

Wisdom-Understanding Why things happen , taking decision

4Q. What is Data in the pyramid?

Ans. Data means raw facts or numbers without meaning.

Example: "100, Maths".

5Q. What is Information in the pyramid?

Ans. Information = Data+context (Answers: Who?, What?, Where?, When?)

Information is data that is processed and has meaning.

Example: "Sourabh scored 100 marks in Maths".

6Q. What is Knowledge in the pyramid?

Ans. Knowledge = Information+interpretation (Answers: How?)

Knowledge is using information with experience and understanding.

Example: "Sourabh is good at Maths".

7Q. What is Wisdom in the pyramid?

Ans. Wisdom= Knowledge+judgement (Answers: Why?)

Wisdom is using knowledge to make the best decisions.

Example: "Sourabh should choose a career in Engineering because of strong Maths skills".

Level	Definition	Formula	Answers to...	Example
Data	Raw facts or numbers without meaning	–	Nothing	"100, Maths"
Information	Data that is processed and has meaning	Data + Context	Who?, What?, Where?, When?	"Sourabh scored 100 marks in Maths"
Knowledge	Using information with experience and understanding	Information + Interpretation	How?	"Sourabh is good at Maths"
Wisdom	Using knowledge to make the best decisions	Knowledge + Judgement	Why?	"Sourabh should choose a career in Engineering because of strong Maths skills"

8Q. What do you mean by Data Literacy Framework?

Ans. A Data Literacy Framework is a structured plan that helps people learn how to understand, use, and communicate data effectively.

It's like a road-map that guides organizations like schools, companies or individuals in building the skills needed to work with data confidently and responsibly.

9Q. What are the steps of Data Literacy Framework?

Ans. Plan → Communicate → Assess → Develop Culture → Learn → Evaluate

10Q. Explain the steps of Data Literacy Framework.

Ans. The explanation of the Steps of the Data Literacy Process Framework is as follows:

Step	Explanation	Example
Plan	Decide the goal, who will be involved, and the time needed.	A school wants to improve students' use of data in Science projects.
Communicate	Share the goal with everyone and explain why it is important.	Teachers tell students, "We want you to learn how to collect and use data in experiments."
Assess	Check the current skill level of participants.	Teachers give a small quiz to see if students can read graphs and tables.
Develop Culture	Create an environment where data literacy becomes a habit.	Students regularly collect and analyze data in class experiments.
Prescriptive Learning	Provide suitable learning materials and activities for different needs.	Some students watch videos, some read guides, and others practice with spreadsheets.
Evaluate	Measure progress and check if goals are achieved.	At the end of the term, students present a Science project using data charts and teachers give feedback.

11Q. What is the difference between data privacy and data security?

Ans. Difference Between Data Privacy and Data Security

Topic	Data Privacy	Data Security
Meaning	Keeping personal information private and not misused	Protecting data from being stolen, changed, or lost
Focus	Who can see or use our data	How our data is protected from hackers or damage
Example	Choosing who can see our Instagram profile	Using a strong password to keep our account safe

12Q. What are the various types of data security controls?

Ans. The different types of data security controls are as follows:

1. **Strong Passwords** – A secret code that protects your account from others.
2. **Multi-factor Authentication (MFA)** – Using two or more steps to prove it's really us.
3. **Firewall** – A digital wall that blocks harmful traffic from entering your computer.
4. **Data Masking** – Hiding real data by replacing it with fake but similar-looking data.
5. **Encryption** – Locking data so only someone with the key can read it.

13Q. What is multi-factor authentication and why it is important?

Ans.

- Multi-Factor Authentication (MFA) is a security method where we **use two or more proofs** (like a password + OTP on phone) to log in.
- It is important because it gives **extra protection** – even if someone steals our password, they cannot access our account without the second proof.
- Example: Logging into Gmail with our password **and** an OTP sent to our mobile.

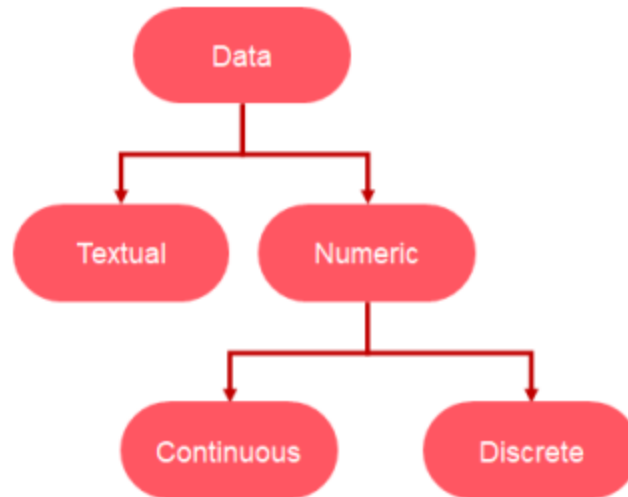
14Q. Discuss the key aspects of Data privacy and their significance in today's digital world.

Ans.

Aspect	Significance in Today's Digital World
Consent	Ensures people control their own data and avoid misuse a when data is collected and used only with the <u>person's permission</u> .
Transparency	Builds trust because <u>people know how their data is being used</u> .
Data Minimization	Reduces risk of data leaks by <u>collecting only what is necessary</u> .
Right to be Forgotten	<u>Allows people to remove old or unnecessary data</u> to protect their privacy.

15Q. What are the different types of Data?

Ans. The data can be categorised into different ways one of them as Textual and Numerical Data-

**16Q. Write the difference between Textual and Numerical Data?**

Ans.

Textual Data (Qualitative Data)	Numeric Data (Quantitative Data)
It is made up of words and phrases	It is made up of numbers
It is used for Natural Language Processing (NLP)	It is used for statistical data
Search queries on the internet are an example of textual data	Any measurements, readings, or values would count as numeric data
Example: “Which is a good park nearby?”	Example: Cricket Score, Restaurant Bill

17Q. How can you further classify Numerical Data?

Ans. Numeric Data is further classified as: Continuous Data and Discrete Data

Continuous Data	Discrete Data
Continuous data is numeric data that can take any value	Discrete data is numeric data that contains only whole numbers and cannot be fractional
E.g., height, weight, temperature, voltage	E.g. the number of students in the class

18Q. What are the various Data Acquisition Techniques?

Ans.- Data Acquisition refers to the procedure of gathering data.

The process typically comprises three key steps:-



19Q. What do you mean by Web Scraping?

Ans. Web Scraping means data acquisition from websites.

Data acquisition from websites

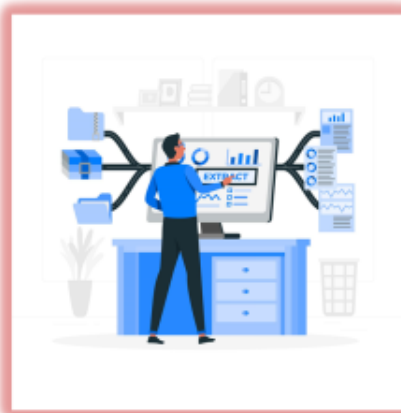


1

The process of **collecting data** from websites using software is called **Web Scraping**

3

While web scraping is not illegal, using data **without permission** is illegal



2

There are different **tools** that can help us collect data from websites

4

During data acquisition, we need to make sure that the data source **allows** data scraping

20Q. What is the difference between Primary Data and Secondary Data?

Ans.

Criteria	Primary Data	Secondary Data
Source	Collected <u>first hand</u> by researcher	<u>Collected by others</u>
Purpose	Specific to current research	May have different original purpose
Time	Takes longer to collect	Quickly available
Cost	More expensive	Often less expensive
Control	Full Control over collection process	No control over collection process
Accuracy	More accurate for specific needs	May be less accurate for specific needs
Freshness	Up to date	May be outdated
Example	A researcher wants to <u>study the eating habits</u> of college students. They create a <u>survey</u> and distribute it to 500 students on campus asking about their daily food choices, meal times, and snacking habits	A researcher studying <u>trends in population growth</u> uses census data which is <u>collected by the government. over the past 50 years</u> , rather than conducting their own population count.

21Q. What are the various sources for acquiring data?

Ans. Various Sources for Acquiring Data:

- **Primary Data Sources** — Some of the sources for primary data include surveys, interviews, experiments, etc.
- **Secondary Data Sources**—Secondary data collection obtains information from external sources like Government Publications, Research reports and Online database etc. Some sources for secondary data collection include: Kaggle, .gov datasets, Google Dataset Search, UCI



22Q. What are the three primary factors that determines the usability of data?

Ans. The three factors are- Structure, Cleanliness and Accuracy.

1. **Structure**- Defines how data is stored. The structured data is considered as more usable than unstructured data.

2. **Cleanliness**- Clean data is free from duplicates, missing values, outliers, and other anomalies that may affect its reliability and usefulness for analysis.

3. **Accuracy**- Accuracy indicates how well the data matches real-world values.

23Q. What do you mean by the features of the data. What are its types?

Ans. Data features are the characteristics or properties of the data.

For example, in a table of student records, features could include things like the student's name, age, or grade.

There are two types of data features: independent and dependent.

Independent features are the input to the model—they're the information we provide to make predictions.

Dependent features, on the other hand, are the outputs or results of the model—they're what we're trying to predict.

24Q. Write the stages of data processing.



Ans.

Stage	Acquire	Process	Analyze	Interpret	Present
Meaning	Collect data from various sources	Data is processed to derive meaningful information	Draw conclusion	Explain what these findings/ conclusions mean	Organize, and group ideas and evidence in a logical way.
Example	A school collects students' marks	Calculates totals	Finds class average	Understands which subjects are weak	Shows report card

25Q. How to Collect and Analyze Qualitative and Quantitative Data?

Ans.

Type of Data	Collect	Analyse
Qualitative Data	Record keeping Observations Case Studies Focus groups Longitudinal Studies One-to-one interviews	5 Steps to Qualitative Data Analysis 1. Collect Data 2. Organize 3. Set a code to the Data Collected 4. Analyze your data 5. Reporting
Quantitative Data	Interviews Polls Observations Longitudinal Studies Survey	4 Steps to Quantitative Data Analysis 1. Relate measurement scales with variables 2. Connect descriptive statistics with data 3. Decide a measurement scale 4. Represent data in an appropriate format.

26Q. What is a longitudinal study?

Ans. A longitudinal study collects data from the same source repeatedly over a long period to identify trends and patterns.

27Q. How to interpret the data

Ans:-Based on the two types of data, there are two ways to interpret data-

1.Qualitative Data Interpretation :

It tells us about the emotions and feelings of people.

It is focused on insights and motivations of people.

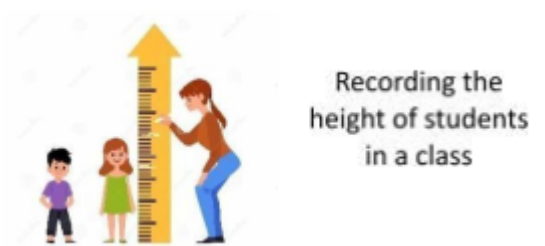
For example – Reviews by customer.

**2. Quantitative Data Interpretation:**

It is made on numerical data.

It helps us answer questions like “when,” “how many,” and “how often”.

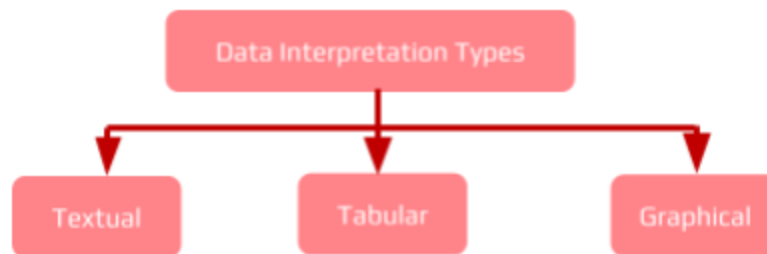
For example – (how many) numbers of likes on the Instagram post, number of visits on website, recording height of students in class.



Qualitative Data Interpretation	Quantitative Data Interpretation
Categorical	Numerical
Provides insights into feelings and emotions	Provides insights into quantity
Answers how and why	Answers when, how many or how often
Methods – Interviews, Focus Groups	Methods – Assessment, Tests, Polls, Surveys
Example question – Why do students like attending online classes?	Example question – How many students like attending online classes?

28Q. What are the different types of data interpretation?

Ans. There are three ways in which data can be presented-



Textual DI

- The data is mentioned in the text form, usually in a paragraph.

Example:

In the Science Olympiad class of 45 Students, 3 students obtained the perfect score of 50. 10 students got a score of 45 and above, 15 students got a score of 40 and above, 8 students got a score of 30 and above, 6 students got a score of 20 and above and 3 got 19 and below.

More than 60% of students scored more than 80% Marks in Olympiad!

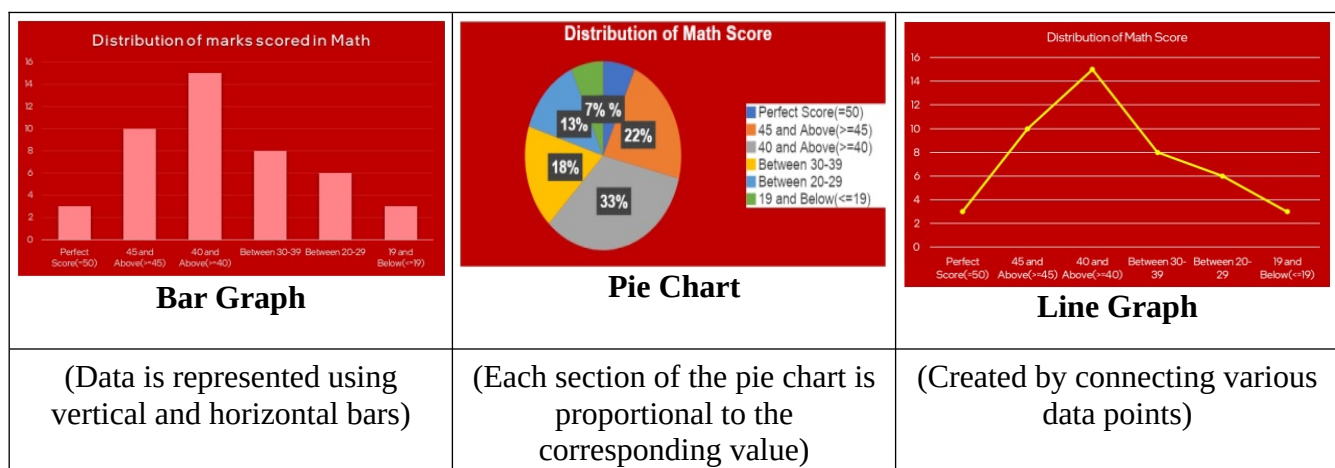
Tabular DI

- Data is represented systematically in the form of rows and columns.

Student Performance in Science Olympiad	
Marks Range	No. of Students
Below 19	3
20-29	6
30-39	8
40-49	25
Perfect 50	3

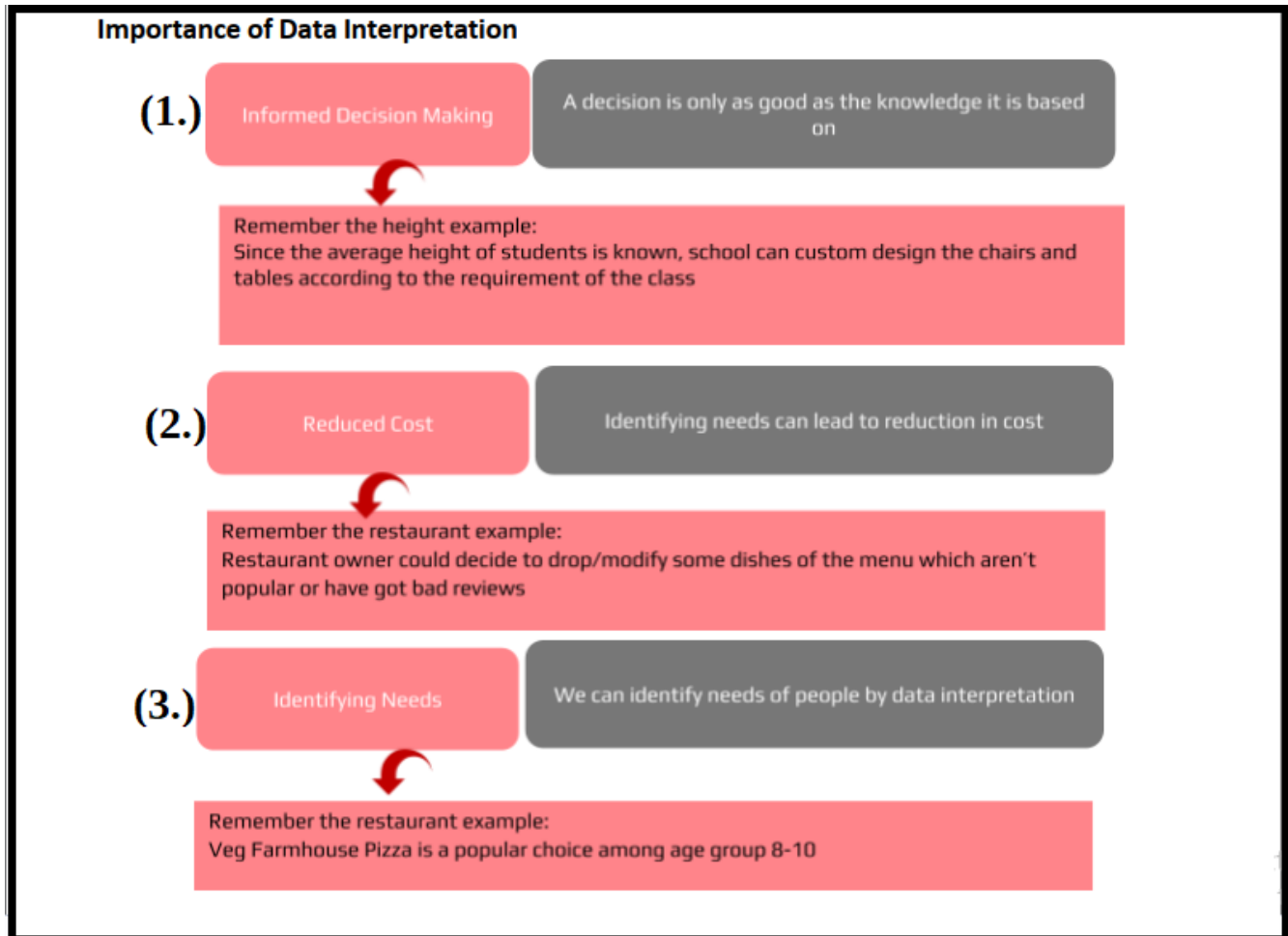
Graphical DI

- Data is represented in the form of graphs and charts like Bar graph, pie chart and line graphs etc.



29Q. What is the importance of data interpretation?

Ans.



30Q. Name the software which make it easier for us to present data.

Ans. Tableau

Q. Why is using 'https' in URLs recommended for cybersecurity?

Ans. 'https' indicates that the website uses a secure connection, protecting data transmission.

Q. Name one best practice for cybersecurity.

Ans. Use strong, unique passwords and enable two-factor authentication for accounts.

Q. What are two examples of quantitative data?

Ans. Examples include numerical measurements like temperature readings and the number of students in a class.

Q. Why is data cleaning important?

Ans. Data cleaning removes duplicates, errors, and inconsistencies, ensuring data reliability and accuracy for analysis.

Q. What is continuous data?

Ans. Continuous data can take any value within a range, such as temperature or height.

Q. What is data augmentation?

Ans. Data augmentation involves increasing the amount of data by making modified copies of existing data.

Q. What is the purpose of data interpretation?

Ans. The purpose of data interpretation is to make sense of processed data and draw meaningful conclusions.

Q. Name one method of qualitative data collection.

Ans. Examples include observations, case studies, or focus groups.

Q. What does a pie chart represent?

Ans. A pie chart represents data as parts of a whole, with each section proportional to its value.

Q. What is meant by 'data acquisition'?

Ans. Data acquisition refers to the process of gathering data from various sources.

Q. What is an independent feature in an AI model?

Ans. An independent feature is an input variable used to make predictions in the model.

Q. What kind of insights can qualitative data provide?

Ans. Qualitative data provides insights into emotions, motivations, and detailed descriptions.

Q. What is data processing?

Ans. Data processing involves transforming raw data into meaningful information through various operations.

Q. What does a bar graph display?

Ans. A bar graph displays data using vertical or horizontal bars to compare different categories.

Q1. _____ means knowing how to understand, work with and talks about data.

- a. Data Relation
- b. Data literacy
- c. Data Understanding
- d. Data study

Ans. b. Data literacy

Q2. Data Literacy means _____

- a. to understand data
- b. to interpret data
- c. to communicate with data
- d. All of the above

Ans. d. All of the above

Q3. Ability to read, comprehend and communicate in a language is known as _____

- a. Literacy
- b. Communication
- c. Literature
- d. Data Literature

Ans. a. Literacy

Q4. Data Pyramid is made of _____ stages of working with data.

- a. One
- b. Two
- c. Three
- d. Four

Ans. d. Four

Q5. Processed data is called _____

- a. Information
- b. Knowledge
- c. Wisdom
- d. Complete data

Ans. a. Information

Q6. _____ allows us to understand why things are happening in a particular way.

- a. Data
- b. Wisdom
- c. Knowledge
- d. Information

Ans. b. Wisdom

Q7. Data literacy is essential because it enables individuals _____

- a. to make informed decisions
- b. to think critically

c. to solve problems and innovate

d. All of the above

Ans. d. All of the above

Q8. Every data tells a story. A person who can interact with data to understand the world around them is _____

a. Data Educator

b. Data Professor

c. Data Literate

d. Data Story Teller

Ans. c. Data Literate

Q9. Aman is a data literate and now he want to buy mobile. Which of the following things he will do while doing online shopping?

a. Filter the category as per the requirement

b. Check the user ratings of the products

c. Check for specific requirements in the product

d. All of the above

Ans. d. All of the above

Q10. Data literacy framework is an iterative process.(True/False)

a. True

b. False

Ans. a. True

Q11. _____ provides guidance on using data efficiently and with all levels of awareness.

a. Data Literacy

b. Data Literature

c. Data Literacy Framework

d. Data Framework

Ans. c. Data Literacy Framework

Q12. Data Privacy and Data Security are not different from each other.(True/False)

a. True

b. False

Ans. b. False

Q13. _____ refers to how data is collected, shared and used.

a. Data Privacy

b. Data Security

c. Security

d. Data Literacy

Ans. a. Data Privacy

Q14. _____ refers to protecting data from hackers who might want to misuse it.

- a. Data Privacy
- b. Data Security
- c. Security
- d. Data Literacy

Ans. b. Data Security

Q15. Proper handling of sensitive data such as certain financial data and intellectual property data as well as protecting the confidentiality and immutability of the data is _____

- a. Data Security
- b. Data Management
- c. Data Protection
- d. Data Privacy

Ans. d. Data Privacy

Q16. Which of the following may compromise our data privacy?

- a. Downloading an unverified mobile application.
- b. Accepting the terms of service without reading.
- c. Both of the above
- d. None of the above

Ans. c. Both of the above

Q17. _____ is the practice of protecting digital information from unauthorized access, corruption, or theft throughout its entire life cycle.

- a. Data Breach
- b. Data Authentication
- c. Data Privacy
- d. Data Security

Ans. d. Data Security

Q18. The most possible reasons why data security is more important are _____

- a. Cyber-attacks
- b. Fast-technological changes will boom cyber attacks
- c. Newer Hacking Methods
- d. All of the above

Ans. d. All of the above

Q19. _____ involves protecting computers, servers, mobile devices, electronic systems, networks, and data from harmful attacks.

- a. Cyber Threat
- b. Cyber Security
- c. Cyber Crime
- d. Cyber Cafe

Ans. b. Cyber Security

Q20. What we can do to enhance cyber security?

- a. Use strong, unique passwords
- b. Activate Two-Factor Authentication (2FA) for added security
- c. Download software from trusted sources
- d. All of the above

Ans. d. All of the above

Q21. We should avoid sharing _____ to prevent Cyber attack?

- a. personal hobbies
- b. personal info like real name or phone number
- c. personal favorite book
- d. personal favorite food

Ans. b. personal info like real name or phone number

Q22. Cultivating Data Literacy means _____

- a. Utilize vocabulary and analytical skills
- b. Acquire, develop, and improve data literacy skills
- c. Develop skills in statistical methodologies
- d. Develop skills in Math

Ans. b. Acquire, develop, and improve data literacy skills

Q23. _____ is the practice of protecting digital information from unauthorized access, corruption, or theft throughout its entire lifecycle.

- a. Data Security
- b. Data Literacy
- c. Data Privacy
- d. Data Acquisition

Ans. a. Data Security

Q24. Which of the following is serving as the foundation of AI?

- a. Data
- b. Humanity
- c. Ethics
- d. Morals

Ans. a. Data

Q25. Two common type of data used in AI is _____

- a. Boolean and Date
- b. Textual and Date
- c. Textual and Numeric
- d. Boolean and Numeric

Ans. c. Textual and Numeric

Q26. Numeric Data used in AI is further classified as _____

- a. Continuous Data and Discrete Data
- b. Qualitative Data and Quantitative Data
- c. Good Data and Bad Data
- d. Continuous Data and Quantitative Data

Ans. a. Continuous Data and Discrete Data

Q27. Textual Data is also known as _____

- a. Boolean Data
- b. Quantitative Data
- c. Qualitative Data
- d. Numeric Data

Ans. c. Qualitative Data

Q28. Numeric Data is also known as _____

- a. Boolean Data
- b. Quantitative Data
- c. Qualitative Data
- d. Textual Data

Ans. b. Quantitative Data

Q29. _____ data is used for Natural Language Processing.

- a. Numeric Data
- b. Textual Data
- c. Date Data
- d. Audio and Visual Data

Ans. b. Textual Data

Q30. _____ data is used for Statistical Data.

- a. Numeric Data
- b. Textual Data
- c. Date Data
- d. Audio and Visual Data

Ans. a. Numeric Data

Q31. Which of the following is an example of Qualitative Data?

- a. Which is a good park nearby?
- b. Vegetarian restaurant near me
- c. Both of the above
- d. Restaurant bill

Ans. c. Both of the above

Q32. Which of the following is an example of Quantitative Data?

- a. Restaurant bill
- b. Cricket Score
- c. Height of a student
- d. All of the above

Ans. All of the above

Q33. Which of the followings are an example of continuous data?

- a. Height, Weight and temperature
- b. Mass, Length and Number of students in class
- c. Temperature, Number of students in class, Percentage in Annual Exam
- d. Number of students in class and Height of a student

Ans. a. Height, Weight and temperature

Q34. _____ is numeric data that contains only whole numbers and cannot be fractional.

- a. Numerical Data
- b. Quantitative Data
- c. Continuous Data
- d. Discrete data

Ans. d. Discrete data

Q35. Visual data like images or videos are used in _____ domain of AI.

- a. NLP
- b. CV
- c. SD
- d. None of the above

Ans. b. CV

Q36. Numeric data like tables or excel sheets are used in _____ domain of AI.

- a. NLP
- b. CV
- c. SD
- d. None of the above

Ans. c. SD

Q37. What type of data is used in NLP?

- a. Images or Videos
- b. Documents or PDF files
- c. Tables or Excel Sheets
- d. None of the above

Ans. b. Documents or PDF files

Q38. _____ refers to the procedure of gathering data

- a. Data Acquisition
- b. Acquiring Data
- c. Both of the above
- d. None of the above

Ans. c. Both of the above

Q39. Data Acquisition typically comprises _____ key steps.

- a. One
- b. Two
- c. Three
- d. Four

Ans. c. Three

Q40. Which of the following is not the key step of Data Acquisition?

- a. Data Discovery
- b. Data sets
- c. Data Augmentation
- d. Data Generation

Ans. b. Data sets

Q41. Aman wants to create a CV model for a self-driving car. For this he require the _____

- a. Pictures of roads
- b. Pictures of different vehicles on road
- c. Pictures of different objects on road
- d. All of the above

Ans. d. All of the above

Q42. Searching and downloading data from the internet for AI model is called _____

- a. Data Base
- b. Data Generation
- c. Data Augmentation
- d. Data Discovery

Ans. d. Data Discovery

Q43. _____ means increasing the amount of data by adding copies of existing data with small changes.

- a. Data Base
- b. Data Generation
- c. Data Augmentation
- d. Data Discovery

Ans. c. Data Augmentation

Q44. In which of the following steps of Data Acquisition, New data is added by slightly changing the existing data.

- a. Data Base
- b. Data Generation
- c. Data Augmentation
- d. Data Discovery

Ans. c. Data Augmentation

Q45. Recording temperature readings of a building is an example of _____

- a. Data Base
- b. Data Generation
- c. Data Augmentation
- d. Data Discovery

Ans. b. Data Generation

Q46. Sources for Acquiring Data is _____

- a. Surveys
- b. Interviews
- c. Experiments
- d. All of the above

Ans. d. All of the above

Q47. _____ is an online community of data scientists where we can access different types of data.

- a. Google
- b. Kaggle
- c. Datasets
- d. UCI

Ans. b. Kaggle

Q48. _____ is a collection of databases, domain theories and data generators in collaboration with the university of Massachusetts.

- a. Google
- b. Kaggle
- c. UCI
- d. .gov datasets

Ans. c. UCI

Q49. Which of the following factors should be considered which make Good Data?

- a. Information is well structured.
- b. It should be Consistent
- c. It should be accurate
- d. All of the above

Ans. d. All of the above

Q50. Which of the following factors make Bad Data?

- a. Information is scattered
- b. Contains inaccurate values
- c. Contains duplicate values
- d. All of the above

Ans. d. All of the above

Q51. The process of collecting data from websites using software is called _____

- a. Web Software
- b. Web Scraping
- c. Web Downloading
- d. Web Sharing

Ans. b. Web Scraping

Q52. Which of the following is not the ethical concerns in data acquisition?

- a. Bias
- b. Consent
- c. Accountability
- d. Certainty

Ans. d. Certainty

Q53. Web Scraping is not illegal but using data without permission is illegal.(True/False)

- a. True
- b. False

Ans. a. True

Q54. _____ means to take necessary permissions before collecting an individual data.

- a. Accountability
- b. Bias
- c. Consent
- d. Anonymity

Ans. c. Consent

Q55. _____ ethical concern in data acquisition protects the identity of a person who is the source of data.

- a. Bias
- b. Accountability
- c. Anonymity
- d. Consent

Ans. c. Anonymity

Q56. _____ ethical concern in data acquisition refers to take responsibility for your action in case of misuse of data.

- a. Bias
- b. Accountability
- c. Anonymity
- d. Consent

Ans. b. Accountability

Q57. _____ ethical concern in data acquisition refers to avoid any preferences or partiality in data.

- a. Bias
- b. Accountability
- c. Anonymity
- d. Consent

Ans. a. Bias

Q58. Which of the following is primary factors determining the usability of data?

- a. Structure of data
- b. Cleanliness
- c. Accuracy
- d. All of the above

Ans. d. All of the above

Q59. In which of the following, structure of the stored data can be considered as good structure?

- a. Spreadsheet
- b. Notepad
- c. Paint
- d. Power Point

Ans. a. Spreadsheet

Q60. _____ is free from duplicates, missing values, outliers, and other anomalies.

- a. Plain Data
- b. Clean Data
- c. Clear Data
- d. No Error Data

Ans. b. Clean Data

Q61. _____ are the characteristics or properties of the data.

- a. Data sets
- b. Data features
- c. Data base
- d. Data design

Ans. b. Data features

Q62. Data features of student table is _____

- a. Name
- b. Age
- c. Class
- d. All of the above

Ans. d. All of the above

Q63. In a photo dataset, features might be the colors present in each image.(True/False)

- a. True
- b. False

Ans. a. True

Q64. In AI models, we need _____ types of features.

- a. 1
- b. 2
- c. 3
- d. 4

Ans. b. 2

Q65. _____ are the input to the model-they're the information we provide to make predictions.

- a. Interdependent features
- b. Dependent features
- c. Independent features
- d. Data Features

Ans. c. Independent features

Q66. _____ are the outputs or results of the model, they're what we're trying to predict.

- a. Interdependent features
- b. Dependent features
- c. Independent features
- d. Data Features

Ans. b. Dependent features

Q67. _____ refers to operating on data to produce meaningful information.

- a. Data Processing
- b. Data Interpretation
- c. Data Features
- d. Dependent Features

Ans. a. Data Processing

Q68. _____ refers to analyzing data to arrive at meaningful decisions.

- a. Data Features
- b. Data Processing
- c. Data Interpretation
- d. Data base

Ans. c. Data Interpretation

Q69. The process of making sense out of data that has been processed is called _____

- | | |
|------------------------|--------------------|
| a. Data Features | b. Data Processing |
| c. Data Interpretation | d. Data base |

Ans. c. Data Interpretation

Q70. _____ is to collect data from various data sources.

- | | |
|---------------------|------------------------|
| a. Independent data | b. Acquiring data |
| c. Processing data | d. Data Interpretation |

Ans. b. Acquiring data

Q71. When raw data is collected and processed to derive meaningful information from it is called _____

- | | |
|--------------------|----------------------|
| a. Data Acquiring | b. Data Analysis |
| c. Data Processing | d. Data Presentation |

Ans. c. Data Processing

Q72. _____ is to examine each component of the data in order to draw conclusions.

- | | |
|--------------------|----------------------|
| a. Data Acquiring | b. Data Analysis |
| c. Data Processing | d. Data Presentation |

Ans. b. Data Analysis

Q73. _____ is to be able to explain what these findings/conclusions mean in a given context.

- | | |
|------------------------|----------------------|
| a. Data Acquiring | b. Data Analysis |
| c. Data Interpretation | d. Data Presentation |

Ans. c. Data Interpretation

Q74. There are _____ ways to interpret data.

- | | |
|------|------|
| a. 1 | b. 2 |
| c. 3 | d. 4 |

Ans. b. 2

Q75. Which of the following data interpretation method tells us about the emotions and feelings of people?

- a. Qualitative Data Interpretation
- b. Quantitative Data Interpretation
- c. Both of the above
- d. None of the above

Ans. a. Qualitative Data Interpretation

Q76. Which of the following data interpretation method is focused on insights and motivations of people?

- a. Qualitative Data Interpretation
- b. Quantitative Data Interpretation
- c. Both of the above
- d. None of the above

Ans. a. Qualitative Data Interpretation

Q77. Reviews given by customer about a restaurant or a hotel is an example of _____

- a. Quantitative data
- b. Data Analysis
- c. Data Interpretation
- d. Qualitative data

Ans. d. Qualitative data

Q78. In _____ method, data is collected using a one-to-one interaction.

- a. Case Studies
- b. Longitudinal Studies
- c. Focus groups
- d. One-to-One Interviews

Ans. d. One-to-One Interviews

Q79. Data Collection Methods for Qualitative Data Interpretation is _____

- a. Case studies
- b. Observation
- c. Focus groups
- d. All of the above

Ans. d. All of the above

Q80. A study conducted over a considerable amount of time to identify trends and patterns is called _____

- a. Case studies
- b. Longitudinal studies
- c. Focus group
- d. Observation

Ans. b. Longitudinal studies

Q81. _____ is the first step for Qualitative Data Analysis.

- a. Organize data
- b. Collect data
- c. Analyze your data
- d. Reporting

Ans. b. Collect data

Q82. _____ is the last step for Qualitative Data Analysis.

- a. Organize data
- b. Collect data
- c. Analyze your data
- d. Reporting

Ans. d. Reporting

Q83. Quantitative data interpretation is made on _____

- a. String
- b. Boolean Data
- c. Numerical data
- d. Characters

Ans. c. Numerical data

Q84. Which of the following is an example of Quantitative Data Interpretation?

- a. Number of website visit
- b. Recording height of students in a class.
- c. CGPA of student
- d. All of the above

Ans. d. All of the above

Q85. Which of the following are Data Collection Methods for Quantitative Data Interpretation?

- a. Interview
- b. Observations
- c. Longitudinal Studies
- d. All of the above

Ans. d. All of the above

Q86. A poll is a type of survey that asks simple questions to respondents. Polls are usually limited to _____ question.

- a. One
- b. Two
- c. Five
- d. Infinite

Ans. a. One

Q87 a. Which of the following question is an example of Qualitative data interpretation?

- a. How many students like attending online classes?
- b. Why do students like attending online classes?

- c. How many students like gardening as hobby?
- d. How many students born in the moth of October?.

Ans. b. Why do students like attending online classes?

Q87 b. Which of the following question is an example of Quantitative data interpretation.

- a. How many students like attending online classes?
- b. Why do students like attending online classes?
- c. Name of the students who like gardening as hobby?
- d. Why we should go for morning walk?

Ans. a. How many students like attending online classes?

Q88. Arrange the following steps of Quantitative data interpretation.

1. Decide a measurement scale
2. Relate measurement scales with variables
3. Represent data in an appropriate format
4. Connect descriptive statistics with data

- a. 1>2>3>4
- b. 2>3>1>4
- c. 2>4>1>3
- d. 2>4>3>1

Ans. c. 2>4>1>3

Q89. _____ method is used when the data is not large and can be easily comprehended by reading.

- a. Textual
- b. Tabular
- c. Graphic
- d. None of the above

Ans. a. Textual

Q90. _____ Data interpretation method is not suitable for large data.

- a. Textual
- b. Tabular
- c. Graphic
- d. None of the above

Ans. a. Textual

Q91. In _____ data is represented systematically in the form of rows and columns

- a. Textual DI
- b. Tabular DI
- c. Graphical DI
- d. None of the above

Ans. b. Tabular DI

Q92. Which of the following is used for Graphical DI?

- a. Bar graph
- b. Line graph
- c. Pie Chart
- d. All of the above

Ans. d. All of the above

Q93. In _____, data is represented using vertical and horizontal bars.

- a. Bar graph
- b. Line graph
- c. Pie Chart
- d. All of the above

Ans. a. Bar graph

Q94. In _____ each slice represents the portion of the entire quantity allocated to each category.

- a. Bar graph
- b. Line graph
- c. Pie Chart
- d. All of the above

Ans. c. Pie Chart

Q95. _____ is a circular chart divided into various sections (think of a cake cut into slices)

- a. Bar graph
- b. Line graph
- c. Pie Chart
- d. All of the above

Ans. c. Pie Chart

Q96. A _____ is created by connecting various data points.

- a. Bar graph
- b. Pie Chart
- c. Line graph
- d. Column chart

Ans. c. Line graph

Q97. Which of the following is important feature of Data Interpretation?

- a. Informed Decision Making
- b. Reduced Cost
- c. Identifying Needs
- d. All of the above

Ans. d. All of the above

Q98. Which software is useful for Data Visualization?

- a. DBMS
- b. MS Word
- c. MS Paint
- d. Tableau

Ans. d. Tableau

Q99. Which of the following chart can not be created using Tableau?

- a. Pie Chart
- b. Bubble Chart
- c. Bar Chart
- d. Dot Dot Chart

Ans. d. Dot Dot Chart

Q100. Tableau can quickly transformed our bar chart to a chart of bubbles. (True/False)

- a. True
- b. False

Ans. a. True